

Lab 2: Spark

Part 1:

```
sara@sara-VirtualBox:~/Desktop/SparkWordCount$ hdfs dfs -cat /output/part-* | sort  
(,5)  
(a,1)  
(Alexandria,1)  
(analytics,1)  
(and,1)  
(Apache,1)  
(applications.,1)  
(big,1)  
(Count,1)  
(data,1)  
(engine.,1)  
(examples.,1)  
(fast,1)  
(Hadoop.,1)  
(Hadoop,2)  
(is,3)  
(Java,1)  
(MapReduce,1)  
(of,1)  
(one,1)  
(processes,1)  
(quickly.,1)  
(simplest,1)  
(Spark,6)  
(teaches,1)  
(the,1)  
(to,1)  
(University,1)  
(used,1)  
(Word,1)  
(write,1)
```

Part 2:

1- Parallel Kmeans Spark:

- Clustering accuracy:

```
salma-rabie@salma-rabie-VirtualBox:~/KMeans/Spark$ nano Accuracy.scala
salma-rabie@salma-rabie-VirtualBox:~/KMeans/Spark$ scalac Accuracy.scala
salma-rabie@salma-rabie-VirtualBox:~/KMeans/Spark$ scala Accuracy
```

Cluster Distribution

```
((2,Iris-virginica),44)
((1,Iris-versicolor),46)
((1,Iris-virginica),6)
((0,Iris-setosa),50)
((2,Iris-versicolor),4)
```

Which means:

Cluster	Setosa	Versicolor	Virginica
0	50	0	0
1	0	46	6
2	0	4	44

- Cluster 0 → Setosa
- Cluster 1 → Versicolor
- Cluster 2 → Virginica

Correctly clustered samples = $50 + 46 + 44 = 140$

Total samples = 150

➤ Accuracy = 93.33%

- Run Time:

```
Iteration 1
Centroid 0 = 5.005999999999999,3.4180000000000006,1.464,0.24399999999999999
Centroid 1 = 5.466666666666667,2.538095238095238,3.8476190476190473,1.180952380952381
Centroid 2 = 6.4734177215189845,2.9607594936708863,5.187341772151901,1.8075949367088608

Iteration 2
Centroid 0 = 5.005999999999999,3.4180000000000006,1.464,0.24399999999999999
Centroid 1 = 5.599999999999998,2.6193548387096777,4.0032258064516135,1.2419354838709677
Centroid 2 = 6.55942028985507,2.985507246376812,5.3115942028985526,1.8710144927536227

Iteration 3
Centroid 0 = 5.005999999999999,3.4180000000000006,1.464,0.24399999999999999
Centroid 1 = 5.672222222222221,2.6722222222222216,4.075,1.2638888888888888
Centroid 2 = 6.593749999999997,2.9843750000000004,5.3734375000000005,1.9078124999999995

Iteration 4
Centroid 0 = 5.005999999999999,3.4180000000000006,1.464,0.24399999999999999
Centroid 1 = 5.712499999999999,2.6774999999999993,4.1425,1.295
Centroid 2 = 6.628333333333329,3.0016666666666666,5.414999999999998,1.9299999999999995

Iteration 5
Centroid 0 = 5.005999999999999,3.4180000000000006,1.464,0.24399999999999999
Centroid 1 = 5.786666666666666,2.702222222222223,4.195555555555556,1.322222222222222
Centroid 2 = 6.650909090909089,3.0109090909090903,5.4872727272727255,1.9654545454545451

Runtime = 3.996787969 seconds
```

2- Unparallel Kmeans:

- Run Time:

```
salma-rabie@salma-rabie-VirtualBox:~/KMeans/Sequential$ nano SequentialKMeans.scala
salma-rabie@salma-rabie-VirtualBox:~/KMeans/Sequential$ scalac SequentialKMeans.scala
salma-rabie@salma-rabie-VirtualBox:~/KMeans/Sequential$ scala SequentialKMeans \
../Spark/iris.data

Iteration 1
Centroid 0 = 5.005999999999999,3.4180000000000006,1.464,0.24399999999999999
Centroid 1 = 5.466666666666667,2.538095238095238,3.8476190476190473,1.180952380952381
Centroid 2 = 6.4734177215189845,2.9607594936708863,5.187341772151901,1.8075949367088608

Iteration 2
Centroid 0 = 5.005999999999999,3.4180000000000006,1.464,0.24399999999999999
Centroid 1 = 5.599999999999998,2.6193548387096777,4.0032258064516135,1.2419354838709677
Centroid 2 = 6.55942028985507,2.985507246376812,5.3115942028985526,1.8710144927536227

Iteration 3
Centroid 0 = 5.005999999999999,3.4180000000000006,1.464,0.24399999999999999
Centroid 1 = 5.672222222222221,2.672222222222216,4.075,1.2638888888888888
Centroid 2 = 6.593749999999997,2.9843750000000004,5.3734375000000005,1.9078124999999995

Iteration 4
Centroid 0 = 5.005999999999999,3.4180000000000006,1.464,0.24399999999999999
Centroid 1 = 5.712499999999999,2.6774999999999993,4.1425,1.295
Centroid 2 = 6.628333333333329,3.0016666666666666,5.414999999999998,1.9299999999999995

Iteration 5
Centroid 0 = 5.005999999999999,3.4180000000000006,1.464,0.24399999999999999
Centroid 1 = 5.786666666666666,2.702222222222223,4.195555555555556,1.322222222222222
Centroid 2 = 6.650909090909089,3.0109090909090903,5.4872727272727255,1.9654545454545451

Runtime = 0.089070536 seconds
```

3- Parallel Kmeans Haboop:

- Run Time:

Iteration 1

```
0      5.0060000000000001,3.4180000000000006,1.4640000000000004,0.244
1      5.466666666666667,2.538095238095238,3.847619047619047,1.1809523809523812
2      6.473417721518987,2.9607594936708863,5.187341772151897,1.8075949367088608
```

Iteration 2

```
0      5.005999999999999,3.418,1.4639999999999997,0.24399999999999999
1      5.6,2.619354838709677,4.0032258064516135,1.241935483870968
2      6.559420289855073,2.9855072463768115,5.31159420289855,1.8710144927536232
```

Iteration 3

```
0      5.005999999999999,3.418,1.4639999999999997,0.24399999999999999
1      5.672222222222221,2.6722222222222225,4.075,1.2638888888888886
2      6.593749999999997,2.984375,5.3734375000000002,1.9078124999999997
```

Iteration 4

```
0      5.0059999999999985,3.418,1.4639999999999997,0.24399999999999994
1      5.712499999999999,2.6775,4.1425,1.295
2      6.6283333333333329,3.0016666666666666,5.415,1.9299999999999997
```

Iteration 5

```
0      5.0059999999999985,3.4179999999999997,1.4639999999999997,0.24399999999999994
1      5.786666666666665,2.702222222222223,4.1955555555555556,1.3222222222222222
2      6.650909090909089,3.0109090909090903,5.487272727272726,1.9654545454545451
```

Total Runtime = 150.517972339 seconds

Final Centroids:

```
Cluster 0 = 5.0059999999999985,3.4179999999999997,1.4639999999999997,0.24399999999999999
Cluster 1 = 5.786666666666665,2.702222222222223,4.1955555555555556,1.3222222222222222
Cluster 2 = 6.650909090909089,3.0109090909090903,5.487272727272726,1.9654545454545451
```

Comparison

Version	Runtime
Sequential	0.089
Hadoop MapReduce	150.51
Spark	3.99

Although Hadoop and Spark provide parallel processing capabilities, the Iris dataset contains only 150 samples. Consequently, the sequential implementation achieved the shortest runtime, while Hadoop incurred the largest overhead due to job scheduling and disk I/O. Spark performed better than Hadoop because it keeps data in memory between iterations.

